Problem M3: Isospin and NN Elastic Cross Sections

Question → What It Wants → Solution

Midterm Preparation Notes

The Question

M3 (verbatim)

"Similar to the example in Griffiths regarding the ratio of cross sections for $N+N \to d+\pi$, use isospin eigenstate kets to motivate why it is expected that the total cross section ratios for σ_{p+p}, σ_{p+n}, & (and σ_{n+n}) (e.g. pp/np) are all one, i.e. all cross sections should be considered equal for elastic scattering in some energy regime, like high but not too high energy."

What It Wants — Broken Down

1. **Review** the Griffiths example ($NN \to d+\pi$) to understand the method: decompose initial/final states into isospin eigenstates, use Clebsch–Gordan coefficients, extract amplitude ratios.
2. **Apply the same method** to $NN \to NN$ elastic scattering: decompose pp, pn, nn into isospin eigenstates.
3. **Write each elastic cross section** in terms of the two independent isospin amplitudes $\mathcal{A}_0$ (for $I = 0$) and $\mathcal{A}_1$ (for $I = 1$).
4. **Argue physically** why at "high but not too high" energies, $\mathcal{A}_0 \approx \mathcal{A}_1$, and show this makes all three cross sections equal.

1 Background: The Griffiths Example ($NN \to d + \pi$)

The Method (Griffiths §4.3)

The strong interaction conserves isospin. So if you decompose initial and final states into definite-I eigenstates, only the same-I components can connect. Each isospin channel has its own amplitude. Cross sections go as |amplitude|^2, and the Clebsch–Gordan coefficients determine how much of each channel contributes.

Quick review of $NN \to d + \pi$

The deuteron has $I = 0$. The pion has $I = 1$. So the final state is:

$$|d + \pi\rangle = |I_d = 0\rangle \otimes |I_\pi = 1, I_{3,\pi}\rangle = |I_{\text{tot}} = 1, I_3 = I_{3,\pi}\rangle \tag{1}$$

The final state is **pure** $I = 1$. Only the $I = 1$ component of the initial NN state contributes.

The three reactions and their initial-state decompositions:

$$(a) \quad p + p \to d + \pi^+ : \qquad |pp\rangle = |1, 1\rangle \quad (\text{pure } I = 1) \tag{2}$$

$$(b) \quad p + n \to d + \pi^0 : \qquad |pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle \tag{3}$$

$$(c) \quad n + n \to d + \pi^- : \qquad |nn\rangle = |1, -1\rangle \quad (\text{pure } I = 1) \tag{4}$$

Only the $I = 1$ piece of the initial state connects to the $I = 1$ final state. The amplitudes are:

$$\mathcal{M}_a = \langle 1, 1|\hat{T}|1, 1\rangle = \mathcal{A}_1 \tag{5}$$

$$\mathcal{M}_b = \langle 1, 0|\hat{T}|\frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 0\rangle)\rangle = \frac{1}{\sqrt{2}}\mathcal{A}_1 \tag{6}$$

$$\mathcal{M}_c = \langle 1, -1|\hat{T}|1, -1\rangle = \mathcal{A}_1 \tag{7}$$

Cross sections $\propto |\mathcal{M}|^2$:

$$\sigma_a : \sigma_b : \sigma_c = |\mathcal{A}_1|^2 : \frac{1}{2}|\mathcal{A}_1|^2 : |\mathcal{A}_1|^2 = 2 : 1 : 2 \tag{8}$$

Why this example matters for M3

In $NN \to d\pi$, only **one** isospin channel ($I = 1$) contributes because the final state is pure $I = 1$. This fixes the ratios completely.

For elastic $NN \to NN$, **both** $I = 0$ and $I = 1$ channels contribute. So the ratios depend on the *relative size* of the two amplitudes. The question asks: when are they equal?

2 Isospin Decomposition of NN States

Setup

The nucleon is an isospin-$\frac{1}{2}$ doublet:

$$|p\rangle = \left|\tfrac{1}{2}, +\tfrac{1}{2}\right\rangle, \qquad |n\rangle = \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle \tag{9}$$

Two nucleons: $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$. The Clebsch–Gordan decomposition (identical to spin-$\frac{1}{2}$ addition):

The Two-Nucleon Isospin States

Isotriplet ($I = 1$, symmetric):

$$|1, +1\rangle = |pp\rangle \tag{10}$$

$$|1, \ \ 0\rangle = \frac{1}{\sqrt{2}}\big(|pn\rangle + |np\rangle\big) \tag{11}$$

$$|1, -1\rangle = |nn\rangle \tag{12}$$

Isosinglet ($I = 0$, antisymmetric):

$$|0, 0\rangle = \frac{1}{\sqrt{2}}\big(|pn\rangle - |np\rangle\big) \tag{13}$$

Inverting to express the physical two-particle states:

$$|pp\rangle = |1, +1\rangle \qquad \text{(pure } I = 1\text{)} \tag{14}$$

$$|nn\rangle = |1, -1\rangle \qquad \text{(pure } I = 1\text{)} \tag{15}$$

$$|pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle \qquad \text{(mixture of } I = 0 \text{ and } I = 1\text{)} \tag{16}$$

The key observation

pp and nn are **pure** $I = 1$. Only pn has an $I = 0$ component. This is why pn is special, and why the cross sections aren't *automatically* equal.

3 Elastic Scattering Amplitudes

The two independent amplitudes

Since the strong interaction conserves isospin, the scattering operator $\hat{T}$ is diagonal in I:

$$\langle I', I_3'|\hat{T}|I, I_3\rangle = \delta_{II'}\delta_{I_3 I_3'}\,\mathcal{A}_I \tag{17}$$

There are exactly **two** independent amplitudes: $\mathcal{A}_0$ (for $I = 0$) and $\mathcal{A}_1$ (for $I = 1$).

$pp \to pp$

From (14), $|pp\rangle = |1, +1\rangle$:

$$\mathcal{M}(pp \to pp) = \langle 1, +1|\hat{T}|1, +1\rangle = \mathcal{A}_1 \tag{18}$$

$$\boxed{\sigma_{pp} \propto |\mathcal{A}_1|^2} \tag{19}$$

$nn \to nn$

From (15), $|nn\rangle = |1, -1\rangle$:

$$\mathcal{M}(nn \to nn) = \langle 1, -1|\hat{T}|1, -1\rangle = \mathcal{A}_1 \tag{20}$$

$$\boxed{\sigma_{nn} \propto |\mathcal{A}_1|^2} \tag{21}$$

So $\sigma_{pp} = \sigma_{nn}$ **exactly** by isospin symmetry alone, regardless of the values of $\mathcal{A}_0$ and $\mathcal{A}_1$. This is because both are pure $I = 1$ — they're related by a rotation in isospin space.

$pn \to pn$

From (16), $|pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle$:

$$\mathcal{M}(pn \to pn) = \left(\frac{1}{\sqrt{2}}\langle 1, 0| + \frac{1}{\sqrt{2}}\langle 0, 0|\right)\hat{T}\left(\frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle\right) \tag{22}$$

$$= \frac{1}{2}\langle 1, 0|\hat{T}|1, 0\rangle + \frac{1}{2}\langle 0, 0|\hat{T}|0, 0\rangle + \underbrace{\text{cross terms}}_{=0 \text{ (isospin conserved)}} \tag{23}$$

$$= \frac{1}{2}\mathcal{A}_1 + \frac{1}{2}\mathcal{A}_0 \tag{24}$$

$$\boxed{\sigma_{pn} \propto \left|\frac{1}{2}\mathcal{A}_1 + \frac{1}{2}\mathcal{A}_0\right|^2} \tag{25}$$

General Cross Section Expressions

$$\sigma_{pp} = \sigma_{nn} \propto |\mathcal{A}_1|^2 \tag{26}$$

$$\sigma_{pn} \propto \left|\frac{\mathcal{A}_0 + \mathcal{A}_1}{2}\right|^2 = \frac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2 \tag{27}$$

In general, $\sigma_{pn} \neq \sigma_{pp}$ because pn samples **both** isospin channels.

4 When Are All Three Equal?

The condition

We need $\sigma_{pp} = \sigma_{pn}$, i.e.:

$$|\mathcal{A}_1|^2 = \frac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2 \tag{28}$$

This is satisfied when:

$$\boxed{\mathcal{A}_0 = \mathcal{A}_1} \tag{29}$$

Check: if $\mathcal{A}_0 = \mathcal{A}_1 \equiv \mathcal{A}$, then:

$$\sigma_{pp} \propto |\mathcal{A}|^2 \tag{30}$$

$$\sigma_{nn} \propto |\mathcal{A}|^2 \tag{31}$$

$$\sigma_{pn} \propto \left|\frac{\mathcal{A} + \mathcal{A}}{2}\right|^2 = |\mathcal{A}|^2 \tag{32}$$

All equal. The ratios are $1 : 1 : 1$. ✓

Why $\mathcal{A}_0 \approx \mathcal{A}_1$ at "high but not too high" energies

The Physical Argument

The two isospin amplitudes $\mathcal{A}_0$ and $\mathcal{A}_1$ differ only because the strong interaction can, in principle, depend on the total isospin of the NN system. The question is: when does it **not** care?

At low energies (resonance region): The scattering can proceed through specific resonances that have definite isospin. For example, if an $I = 1$ resonance exists at a particular energy but no $I = 0$ resonance, then $\mathcal{A}_1 \gg \mathcal{A}_0$ at that energy, and the cross sections will **not** be equal. This is analogous to the $\Delta(1232)$ resonance in πN scattering, where $I = 3/2$ dominates and gives $\sigma(\pi^+ p)/\sigma(\pi^- p) = 3$.

At high (but not too high) energies: Above the resonance region, nucleon-nucleon scattering becomes dominated by **diffractive/Pomeron exchange**. The Pomeron carries the quantum numbers of the vacuum: $I = 0$, $C = +1$. Because it is flavor-blind, it scatters $I = 0$ and $I = 1$ NN states with equal strength. Physically, at these energies the nucleons are "black disks" that scatter off each other's geometric cross section, which is the same regardless of whether you're a proton or neutron. The strong force at these scales doesn't resolve the isospin of the system.

Equivalently: at high energies, the scattering probes short distances where QCD is approximately flavor-symmetric (the u and d quark mass difference is negligible compared to the momentum transfer). The interaction becomes **charge-independent**: it doesn't distinguish proton from neutron beyond what isospin already accounts for, and both isospin channels scatter identically.

At very high energies: Electroweak effects, which *do* distinguish proton from neutron (through electric charge), start to become relevant. Coulomb corrections to pp scattering differ from nn and pn. Isospin is no longer an exact symmetry. So the equality breaks down again.

Hence: "high but not too high" = above resonances, below electroweak corrections.

5 Contrast with the $NN \to d + \pi$ Case

It's instructive to see why the two problems give different-looking answers.

	$NN \to d + \pi$	$NN \to NN$ **(elastic)**
Isospin channels	Only $I = 1$ contributes (final state is pure $I = 1$)	Both $I = 0$ and $I = 1$ contribute
Number of amplitudes	1 ($\mathcal{A}_1$ only)	2 ($\mathcal{A}_0$ and $\mathcal{A}_1$)
pp cross section	$\propto \lvert\mathcal{A}_1\rvert^2$	$\propto \lvert\mathcal{A}_1\rvert^2$
pn cross section	$\propto \frac{1}{2}\lvert\mathcal{A}_1\rvert^2$	$\propto \frac{1}{4}\lvert\mathcal{A}_0 + \mathcal{A}_1\rvert^2$
nn cross section	$\propto \lvert\mathcal{A}_1\rvert^2$	$\propto \lvert\mathcal{A}_1\rvert^2$
Ratios	2 : 1 : 2 (exact, from isospin alone)	1 : 1 : 1 only if $\mathcal{A}_0 = \mathcal{A}_1$ (requires physical argument)

The $NN \to d\pi$ ratios are *exact* predictions of isospin symmetry — they hold at any energy. The NN elastic equality is an *approximate* prediction that requires a physical argument about the energy regime.

Summary

M3 Answer

Step 1: Decompose NN states into isospin eigenstates:

$$|pp\rangle = |1,+1\rangle, \qquad |nn\rangle = |1,-1\rangle, \qquad |pn\rangle = \tfrac{1}{\sqrt{2}}|1,0\rangle + \tfrac{1}{\sqrt{2}}|0,0\rangle$$

Step 2: Elastic scattering amplitudes:

$$\mathcal{M}(pp) = \mathcal{M}(nn) = \mathcal{A}_1$$
$$\mathcal{M}(pn) = \tfrac{1}{2}(\mathcal{A}_0 + \mathcal{A}_1)$$

Step 3: Cross sections:

$$\sigma_{pp} = \sigma_{nn} \propto |\mathcal{A}_1|^2$$
$$\sigma_{pn} \propto \tfrac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2$$

Step 4: At energies above the resonance region but below electroweak scales, the interaction is flavor-blind (Pomeron-dominated, charge-independent). Both isospin channels scatter identically: $\mathcal{A}_0 \approx \mathcal{A}_1$.

Conclusion:

$$\boxed{\sigma_{pp} : \sigma_{pn} : \sigma_{nn} = 1 : 1 : 1}$$

The Takeaway

Isospin alone tells you $\sigma_{pp} = \sigma_{nn}$ exactly, but says nothing definitive about σ_{pn} without knowing the ratio $\mathcal{A}_0/\mathcal{A}_1$. The physical argument that scattering becomes isospin-blind at intermediate-to-high energies is what completes the picture and gives you all three equal. This is a beautiful example of how symmetry arguments *constrain* but don't always *determine* — sometimes you need dynamics too.